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Evaluation of 3-year changes in panoramic radiomorphometric indices and fractal dimension values in primary hyperparathyroidism patients

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Abstract

Objective Longitudinal radiographic data on how primary hyperparathyroidism (PHPT) affects the jawbone are scarce. This study aimed to quantify 3-year mandibular changes in PHPT using radiomorphometric indices and fractal dimension (FD) analysis on panoramic radiographs.

Methods A retrospective, age- and gender-matched, 3-year longitudinal comparison was performed using 120 panoramic radiographs from 60 individuals (30 PHPT; 30 controls). Each participant contributed baseline and 3-year images (mean interval: 3.26 ± 0.13 years). Radiomorphometric indices (SI, MI, AI, PI, PMI) and FD from three standardized regions of interest (ROI1: condyle; ROI2: angle; ROI3: mental) were measured. Intra-observer reliability was excellent (ICCs 0.80–0.96). Between-group and within-group comparisons used Mann–Whitney *U* and Wilcoxon signed-rank tests (two-sided, $p < 0.05$).

Results The antegonial index (AI) was lower in PHPT than controls at baseline (3.26 ± 0.43 vs. 3.62 ± 0.81 mm; $p = 0.034$) and remained lower at 3 years (3.24 ± 0.35 vs. 3.55 ± 0.43 mm; $p = 0.050$). Other indices (SI, MI, PI, PMI) showed no between-group differences at either timepoint and no significant within-group change (all $p > 0.05$). FD analysis showed a lower ROI2 (mandibular angle) value in PHPT at baseline ($p = 0.050$) and a significant difference at 3 years ($p < 0.01$); ROI1 and ROI3 did not differ between groups at either timepoint and exhibited no temporal change (all $p > 0.05$).

Conclusions In a 3-year age- and gender-matched comparison, PHPT was associated with consistently lower AI and reduced FD in the mandibular angle, while other indices and regional FDs were largely unchanged. These findings identify panoramic AI and ROI2-FD as adjunctive, non-invasive markers that could prompt biochemical evaluation when clinical suspicion exists; confirmation and clinical integration require larger multicenter prospective studies with concurrent biochemical profiling.

Keywords Primary hyperparathyroidism, Panoramic radiography, Fractal analysis, Morphometric indices

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Introduction

Parathyroid hormone (PTH) is a critical hormone that regulates the homeostasis of free serum calcium and plays a key role in bone remodeling. Hyperparathyroidism develops as a result of excessive PTH secretion and exists in three forms: primary, secondary, and tertiary. According to Fraser et al., primary hyperparathyroidism (PHPT) is the third most common endocrine disorder [1]. This condition typically arises due to causes such as parathyroid adenoma or gland hyperplasia. In recent years, asymptomatic cases have been more frequently diagnosed due to the widespread use of routine biochemical tests such as PTH, serum calcium (Ca), phosphorus (*P*), and alkaline phosphatase (ALP) levels [1–3].

In PHPT, elevated PTH levels accelerate bone resorption by increasing osteoclastic activity. This mechanism occurs through the osteoprotegerin (OPG)–receptor activator of nuclear factor kappa-*B* ligand (RANKL)–receptor activator of nuclear factor kappa-*B* (RANK) system. PTH acts on osteoblasts and osteocytes to enhance RANKL production. This stimulates the differentiation and activity of osteoclast precursors, leading to high bone turnover. Consequently, cortical thinning, disruption of trabecular architecture, and a decrease in bone mineral density occur. These skeletal changes are evident not only in the axial and appendicular skeleton but also in the jawbones [4–6]. The progression of these changes may be influenced by patient age, sex, disease duration, and whether medical or surgical treatment is initiated during follow-up, as such interventions can directly modify bone metabolism.

In the jawbones, radiographic findings associated with PHPT include loss of the lamina dura, a "ground glass" appearance of the trabecular structure, root resorption, and tooth mobility [6–8]. Brown tumors appear as radiolucent areas, more frequently in the posterior mandible [9–12]. High-turnover bone disorders, such as osteitis fibrosa and renal osteodystrophy, may present with similar jawbone manifestations [13, 14]. Clinically, facial asymmetry, jaw pain, tooth mobility, and displacement are among the presenting features [6–15].

Panoramic radiographs are widely used in dentistry because of their low radiation dose, accessibility, and rapid acquisition. They provide a broad view of both jaws, enabling assessment of general bone architecture [16–18]. While cone-beam computed tomography (CBCT) offers superior spatial resolution, panoramic radiographs remain the modality of choice for large-scale and longitudinal studies due to their lower radiation burden, cost-effectiveness, and availability in routine dental practice. However, the lower resolution and potential distortion inherent to panoramic imaging should be acknowledged as limitations, particularly when evaluating fine trabecular details.

Bone quality in panoramic images can be quantified using radiomorphometric indices such as the mandibular cortical index (MCI), mandibular cortical width (MCW), mental index (MI), antegonial index (AI), and panoramic mandibular index (PMI) [19–21]. These indices are commonly used to assess mandibular bone quality in metabolic bone diseases, including PHPT, as they provide quantitative information on both cortical thickness and trabecular integrity across different anatomical regions. AI, for example, reflects cortical morphology in the antegonial region, which may be more susceptible to metabolic bone changes in PHPT due to its biomechanical stress distribution. Changes in bone micromorphology can also be evaluated using fractal dimension (FD) analysis [22]. The concept of fractals, introduced by Mandelbrot, describes structures with complex geometry that cannot be characterized by integer dimensions [23]. In digital radiographic analysis, FD is commonly measured via the box-counting method, which calculates the slope of the logarithmic plot of pixel counts across different grid sizes [24]. Bone's trabecular network exhibits fractal properties, making FD analysis a sensitive tool for detecting subtle microstructural changes that may not be apparent in linear measurements [25]. Given the trabecular disruption in PHPT, FD analysis may reveal quantitative alterations overlooked by radiomorphometric indices.

Recent studies have explored mandibular bone changes in PHPT using these methods [26–29]. Cantürk et al. [27] reported significantly lower FD values in the mandibular angle region of PHPT patients, suggesting reduced trabecular complexity, while Miloğlu et al. [28] found advanced cortical erosion via MCI despite no significant differences in PMI, MI, or MCW. Both studies were cross-sectional, limiting their ability to capture disease progression. Longitudinal studies, particularly those spanning multiple years, are therefore essential to better understand the natural history of jawbone changes in PHPT and to assess how these changes may be modified by patient demographics and treatment status.

This study aimed to fill the gap in the literature by evaluating longitudinal mandibular changes in PHPT patients through fractal analysis and morphometric measurements, based on the hypothesis that these patients would exhibit progressive cortical thinning and reduced trabecular complexity compared to matched controls.

Materials and methods

Study design and sample

This retrospective longitudinal study was conducted in accordance with the principles of the Declaration of

Helsinki and approved by the Ordu University Clinical Research Ethics Committee (Approval No: 2024/178). The required sample size for the study was calculated using a priori power analysis in G*Power software (version 3.1.9.4), taking into account all radiomorphometric indices and fractal analysis variables. The analysis was conducted with 80% statistical power ($1-\beta=0.80$), a 5% Type I error rate ($\alpha=0.05$), and an effect size (Cohen's d) of 0.7 derived from previous studies on metabolic bone disorders. The calculation was based on the Wilcoxon–Mann–Whitney test for comparing two independent groups and determined that a minimum of 28 participants per group would be sufficient.

Panoramic radiographs taken between 2011 and 2024 at the Department of Oral and Maxillofacial Radiology, Ordu University Faculty of Dentistry, were reviewed. The study included 60 individuals, comprising 30 patients with primary hyperparathyroidism (PHPT) and 30 healthy controls, aged between 18 and 78 years. For each participant, two panoramic radiographs taken approximately 3 years apart (mean interval: 3.26 ± 0.134 years) were analyzed, yielding a total of 120 radiographs. Controls were matched to the PHPT group by age and sex using frequency matching, and statistical analysis confirmed no significant differences between the groups ($p > 0.05$).

All PHPT patients had been diagnosed by an endocrinologist before the first radiograph and remained under medical treatment, including calcium-regulating therapies and vitamin D supplementation, throughout the follow-up period. None underwent parathyroidectomy during the study period. Biochemical parameters such as serum PTH, calcium (Ca), phosphorus (*P*), and alkaline phosphatase (ALP) were not available in the retrospective dataset.

Inclusion and exclusion criteria

The diagnosis of PHPT was based on the medical records in patient files. Participants with any other metabolic bone disorder, dental implants, mini-screws, or fixation plates in the mandible, a history of mandibular surgery due to trauma, tumors/cystic pathologies in the mandibular region, or a history of smoking were excluded. Panoramic radiographs with inadequate image quality were also excluded. The control group was selected according to the same exclusion criteria and had no history of metabolic disorders.

Radiographic method

All panoramic radiographs were obtained using the Planmeca Promax 2D (Planmeca Inc., Helsinki, Finland) device under standardized exposure parameters (65 kV, 5 mA, 14 s) recommended by the manufacturer. Patients were positioned according to the device guidelines to minimize distortion.

Fractal analysis

Fractal analysis was performed using ImageJ software (version 1.5e; National Institutes of Health, USA) following the method described by White and Rudolph [30]. A Gaussian filter with a sigma value of 8 was applied to the selected region of interest (ROI), and the filtered image was subtracted from the original. The gray level was standardized to 128, and the image was binarized. Erosion and dilation operations were applied, followed by skeletonization. The fractal dimension (FD) was calculated using the box-counting method with box sizes of 2, 3, 4, 6, 8, 12, and 16 (Fig. 1).

Region of interest selection

Three standardized ROIs, each measuring 32×32 pixels, were evaluated for each radiograph. ROI1 was placed over the mandibular condyle head, ROI2 in the mandibular angle region between the gonion and mandibular canal, and ROI3 was positioned anterior to the mental foramen, at a point corresponding to two-thirds of the distance between the alveolar crest and the mandibular cortex. This placement was performed according to the same anatomical references in all cases, regardless of dentition status. ROIs were centered on the midpoint of the largest possible circle within the anatomical boundaries to ensure consistency (Fig. 2). Left and right

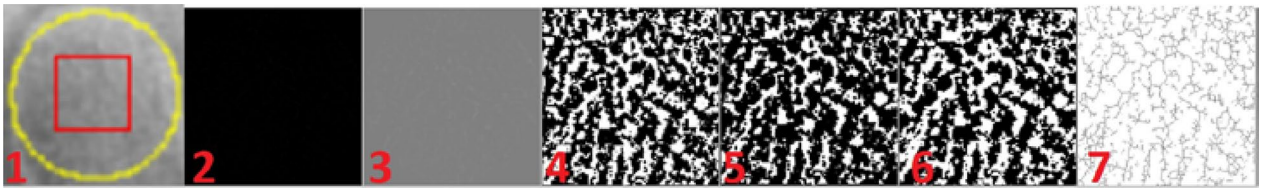

Fig. 1 Steps of fractal dimension analysis: (1) a Gaussian filter with a sigma value of 8 was applied to the region of interest (ROI). (2) The filtered image was subtracted from the original image to enhance structural details. (3) The gray level was adjusted to a fixed value of 128. (4) The image was binarized. (5) The binarized image was subjected to erosion. (6) Dilation was applied to the eroded image. (7) The final skeletonized image was obtained

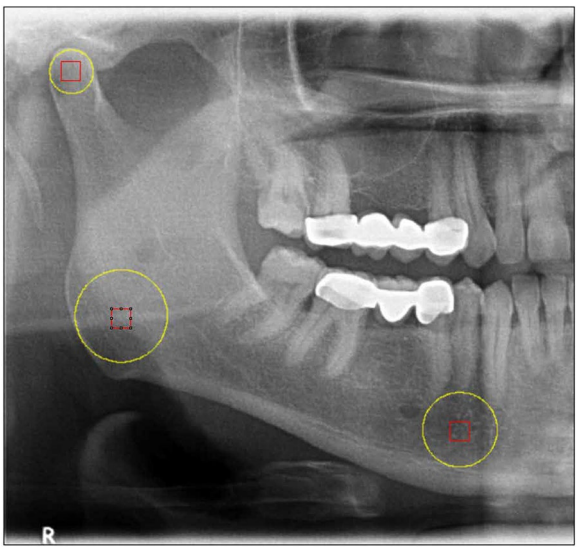

Fig. 2 Regions where fractal dimension analysis was shown on panoramic radiography

sides were analyzed separately, and mean values were calculated. Dentition status (dentate or edentulous) was not used as a criterion for ROI placement. All ROIs were placed according to standardized anatomical landmarks irrespective of missing teeth.

Morphometric indices

Radiomorphometric measurements were performed using Turcasoft Dent software (Turcasoft, Samsun, Türkiye). Five indices were evaluated: symphysis index (SI), antegonial index (AI), molar index (MI), posterior index (PI), and panoramic mandibular index (PMI). SI was measured perpendicular to the inferior mandibular cortex at a point equidistant from the centers of both mental foramina. AI was measured as the cortical thickness at the point where a tangent to the anterior border of the ramus intersected a line along the inferior mandibular border. MI was measured 1 cm distal to the mental foramen, and PI was measured 2.5 cm posterior to the mental foramen, both parallel to the long axis of the mandible and perpendicular to the cortex. PMI was calculated as the ratio of the mental cortical width to the distance between the inferior mandibular cortex and the lower border of the mental foramen (Fig. 3). All measurements were made bilaterally, and average values were used for analysis.

Statistical analysis

All measurements were repeated by the same examiner after 1 month to assess intra-observer reliability. Intraclass correlation coefficients (ICCs) with 95% confidence intervals were calculated and indicated excellent reproducibility. Statistical analyses were performed using IBM SPSS Statistics for Windows (Version 26.0; IBM Corp., Armonk, NY, USA). The Shapiro–Wilk test was used to assess data normality. As variables were not normally distributed, between-group comparisons were performed using the Mann–Whitney *U* test, and within-group changes over time were evaluated with the Wilcoxon signed-rank test. Gender distribution between groups was compared using the Chi-square test. Statistical significance was set at $p < 0.05$.

Results

In this study, a total of 60 individuals, including 30 PHPT patients and 30 healthy controls, were evaluated. The initial mean age of the PHPT group was

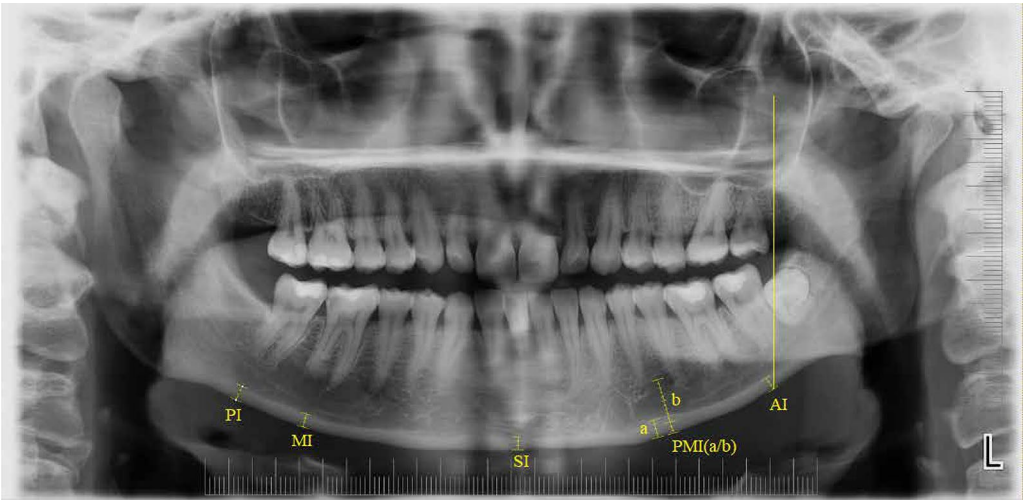

Fig. 3 T The measurements of PI, MI, SI, PMI, and AI on panoramic radiography

Table 1 Distribution of participants in the patient and control groups according to age and gender

	Control	Patient	Total	*P*
Age (years) Mean ± SD	37.11 ± 16.23	36.59 ± 16.50	36.85 ± 16.41	0.872*
Gender	N %	N %	N %	
Male	9 30	9 30	18 30	
Female	21 70	21 70	42 70	1.000**
Total	30 100	30 100	60 100	

Mean: average, SD: standard deviation, * = Mann–Whitney U test,** = Chi-square test p <.05 was considered statistically significant

Table 2 Intraclass correlation coefficients

		ICC	%95 CI	*P**
Initial	SI	0.940	0.813–0.973	**< 0.001**
	MI	0.946	0.886–0.972	**< 0.001**
	PI	0.888	0.548–0.956	**< 0.001**
	AI	0.881	0.800–0.929	**< 0.001**
	PMI	0.946	0.909–0.968	**< 0.001**
	ROI 1	0.934	0.842–0.969	**< 0.001**
	ROI 2	0.912	0.785–0.958	**< 0.001**
	ROI 3	0.927	0.801–0.964	**< 0.001**
Three years	SI	0.898	0.796–0.945	**< 0.001**
	MI	0.955	0.923–0.973	**< 0.001**
	PI	0.797	0.692–0.952	**< 0.001**
	AI	0.823	0.540–0.916	**< 0.001**
	PMI	0.887	0.810–0.932	**< 0.001**
	ROI 1	0.901	0.782–0.947	**< 0.001**
	ROI 2	0.889	0.751–0.938	**< 0.001**
	ROI 3	0.895	0.768–0.942	**< 0.001**

$p < 0.05$ was considered significant and bold values indicate statistical significance, *ICC* intraclass correlation coefficients, *CI* confidence interval, *SI* symphysis index, *MI* mental index, *PI* posterior index, *AI* antegonial index, *PMI* panoramic mandibular index

36.59 ± 16.50 years (range: 18–78), while the control group's mean age was 37.11 ± 16.23 years (range: 18–78). There were no between-group differences in age ($p = 0.872$) or gender distribution ($p = 1.000$) (Table 1).

Radiomorphometric measurements (SI, MI, AI, PI, PMI) were assessed. Intraclass correlation coefficients (ICCs) ranged from 0.797 to 0.955, and all were statistically significant ($p < 0.001$) (Table 2).

At baseline, a between-group difference was detected only for AI (control: 3.62 ± 0.81; PHPT: 3.26 ± 0.43; $p = 0.034$). No between-group differences were observed for SI, MI, PI, or PMI (all $p > 0.05$). At 3-year follow-up, the between-group p-value for AI was 0.050 (control: 3.55 ± 0.43; PHPT: 3.24 ± 0.35), while SI, MI, PI, and PMI remained not significantly different between the groups (all $p > 0.05$). Within-group comparisons across time showed no significant changes for SI, MI, PI, AI, or PMI (all $p > 0.05$) (Table 3).

Fractal analysis showed a between-group p-value of 0.050 for ROI2 at baseline (control: 1.44 ± 0.62; PHPT: 1.40 ± 0.05). At 3 years, a between-group difference was observed for ROI2 ($p < 0.01$; control: 1.45 ± 0.06; PHPT: 1.39 ± 0.04). No between-group differences were found for ROI1 or ROI3 at baseline or at 3 years (all p > 0.05). In the control group, ROI1 was 1.36 ± 0.10 at baseline and 1.35 ± 0.12 at 3 years; ROI3 was 1.34 ± 0.14 at baseline and 1.32 ± 0.17 at 3 years. In the PHPT group, ROI1 was 1.34 ± 0.15 at baseline and 1.34 ± 0.13 at 3 years; ROI3 was 1.31 ± 0.14 at baseline and 1.30 ± 0.11 at 3 years (Table 4).

In gender-stratified analyses of radiomorphometric indices, at baseline no between-group differences were observed among females for SI, MI, PI, or PMI (all $p > 0.05$), whereas AI differed ($p = 0.042$). Among males at baseline, no between-group differences were found for SI, MI, PI, AI, or PMI (all $p > 0.05$). At 3 years, AI differed among females ($p = 0.020$) and among males ($p = 0.040$); PMI also differed among males ($p = 0.036$), while SI, MI, and PI remained not significantly different in both sexes (all $p > 0.05$) (Table 5).

In age-stratified analyses of radiomorphometric indices, at baseline no between-group differences were observed in the < 35 or ≥ 35 age groups for SI, MI, PI, AI, or PMI (all $p > 0.05$). At 3 years, between-group differences were observed in the < 35 group for MI ($p = 0.017$) and AI ($p = 0.040$), while SI, PI, and PMI were not significantly different (all $p > 0.05$). In the ≥ 35 group at 3 years, no between-group differences were detected for SI, MI, PI, AI, or PMI (all $p > 0.05$) (Table 6).

In gender-stratified fractal analysis, at baseline a between-group difference was observed for ROI1 among females ($p = 0.028$), while ROI2 and ROI3 were not significantly different among females or males (all $p > 0.05$). At 3 years, a between-group difference was observed for ROI2 among females ($p = 0.001$); ROI1 and ROI3 were not significantly different among females or males (all $p > 0.05$) (Table 7).

In age-stratified fractal analysis, at baseline, no between-group differences were observed in the < 35 or ≥ 35 groups for ROI1, ROI2, or ROI3 (all $p > 0.05$). At 3 years, between-group differences were observed for ROI2 in both age groups (< 35: $p = 0.002$; ≥ 35: $p = 0.030$). For ROI3, a between-group difference was observed in the < 35 group ($p = 0.040$), while no difference was observed in the ≥ 35 group ($p = 0.599$). ROI1 showed no between-group differences at 3 years in either age group (all $p > 0.05$) (Table 8).

Table 3 Descriptive statistics of SI, MI, PI, AI and PMI values by initial and three years

Initial										Three years								
Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
SI	C	2.57	1.67	4.30	0.60	2.34–2.79	−0.31–0.22	0.06	0.736	2.59	1.84	4.51	0.44	2.43–2.75	−0.29–0.09	0.07	0.300	0.280
	P	2.52	1.77	3.44	0.44	2.35–2.68				2.49	1.69	3.47	0.32	2.38–2.60				0.709
MI	C	3.16	2.09	5.01	0.55	2.95–3.36	−0.35–0.12	0.17	0.346	3.06	2.05	4.78	0.36	2.93–3.19	−0.35–0.05	0.14	0.144	0.202
	P	3.04	2.31	3.74	0.38	2.89–3.18				2.91	2.09	3.71	0.43	2.76–3.06				0.144
PI	C	2.70	2.11	4.47	0.50	2.51–2.88	−0.22–0.20	0.02	0.909	2.60	2.20	4.28	0.30	2.49–2.71	−0.22–0.05	0.02	0.208	0.623
	P	2.69	2.20	3.28	0.30	2.57–2.80				2.69	2.20	3.05	0.24	2.60–2.78				0.180
AI	C	3.62	2.18	5.84	0.81	3.31–3.92	−0.69–−0.02	0.38	**0.034**	3.55	2.12	5.55	0.43	3.40–3.70	−0.49–−0.08	0.35	0.050	0.564
	P	3.26	2.54	4.27	0.43	3.09–3.42				3.24	2.45	4.17	0.35	3.11–3.37				0.746
PMI	C	0.29	0.20	0.45	0.06	0.26–0.31	−0.03–0.01	0.11	0.546	0.29	0.22	0.46	0.04	0.28–0.30	−0.02–0.01	0.05	0.841	0.105
	P	0.28	0.22	0.36	0.04	0.26–0.29				0.28	0.22	0.37	0.04	0.27–0.29				0.825

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* effect size (rank-biserial correlation); *SI* symphysis index; *MI* mental index; *PI* posterior index; *AI* antegonial index; *PMI* panoramic mandibular index; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank test. $p < 0.05$ was considered statistically significant , and bold values indicate statistical significance

Table 4 Descriptive statistics of ROI1, ROI2, and ROI3 values by initial and three years

Initial										Three years								
Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
ROI1	C	1.36	0.99	1.49	0.10	1.32–1.40	−0.09–0.03	0.07	0.381	1.35	1.03	1.52	0.12	1.31–1.39	−0.06–0.06	0.04	0.944	0.279
	P	1.34	0.90	1.56	0.15	1.29–1.39				1.34	1.03	1.56	0.13	1.29–1.39				0.759
ROI2	C	1.44	1.29	1.56	0.62	1.22–1.66	−0.07–−0.01	0.25	0.050	1.45	1.32	1.59	0.06	1.43–1.47	−0.07–−0.02	0.51	. **<0.01**	0.831
	P	1.40	1.28	1.51	0.05	1.38–1.42				1.39	1.32	1.48	0.04	1.38–1.40				0.345
ROI3	C	1.34	0.96	1.51	0.14	1.29–1.39	−0.10–0.04	0.09	0.389	1.32	0.72	1.54	0.17	1.26–1.38	−0.09–0.05	0.07	0.658	0.621
	P	1.31	1.03	1.52	0.14	1.26–1.36				1.30	1.13	1.52	0.11	1.26–1.34				0.874

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* effect size (rank-biserial correlation); *ROI1* condyle region; *ROI2* angular region; *ROI3* mental region; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank Test. $p < 0.05$ was considered statistically significant, and bold values indicate statistical significance

Table 5 Descriptive statistics and within-group comparisons of SI, MI, PI, AI, and PMI values at baseline and 3-year follow-up in female and male participants

	Female										Male								
	Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
Initial	SI	C	2.50	1.91	3.23	0.43	2.10–2.82	−0.29–0.26	0.02	0.937	2.74	1.67	4.30	0.94	1.94–3.67	−0.89–0.60	0.20	0.689	0.346
		P	2.49	1.77	3.44	0.48	2.03–2.89				2.60	2.17	2.87	0.27	2.32–2.83				0.561
	MI	C	3.08	2.35	3.78	0.38	2.79–3.37	−0.30–0.19	0.12	0.683	3.26	2.40	5.01	0.84	2.59–3.62	−0.97–0.38	0.46	0.371	0.563
		P	3.07	2.31	3.74	0.40	2.74–3.42				2.96	2.64	3.46	0.31	2.66–3.27				0.499
	PI	C	2.66	2.16	3.85	0.38	2.40–2.87	−0.13–0.29	0.21	0.474	2.79	2.11	4.47	0.76	2.21–3.11	−0.85–0.33	0.46	0.369	0.525
		P	2.73	2.20	3.28	0.32	2.47–3.00				2.53	2.32	2.84	0.18	2.38–2.67				**0.040**
	AI	C	3.59	2.23	4.99	0.65	3.29–3.89	−0.65–0.04	0.53	**0.042**	3.72	2.18	5.84	1.17	2.86–4.66	−1.42–0.40	0.60	0.248	0.706
		P	3.28	2.54	4.27	0.47	2.94–3.70				3.20	2.65	3.53	0.27	3.05–3.43				0.584
	PMI	C	0.29	0.25	0.36	0.03	0.24–0.34	−0.03–0.03	0.10	0.718	0.28	0.24	0.37	0.04	0.24–0.31	−0.06–0.02	0.47	0.354	0.456
		P	0.29	0.25	0.36	0.03	0.27–0.33				0.26	0.22	0.33	0.03	0.23–0.29				**0.020**
Three years later	SI	C	2.57	1.84	3.51	0.48	2.10–2.96	−0.34–0.14	0.25	0.401	2.67	2.24	2.94	0.27	2.39–2.90	−0.45–0.25	0.31	0.542	0.561
		P	2.46	2.03	2.97	0.29	2.26–2.72				2.56	2.05	2.94	0.38	2.23–2.93				0.492
	MI	C	3.08	2.35	3.78	0.38	2.78–3.29	−0.36–0.12	0.30	0.325	3.00	2.68	3.50	0.32	2.70–3.31	−0.67–0.18	0.60	0.248	0.613
		P	2.96	2.21	3.71	0.41	2.64–3.17				2.76	2.09	3.64	0.46	2.55–3.09				0.312
	PI	C	2.74	2.20	3.28	0.32	2.47–3.01	−0.27–0.06	0.37	0.219	2.53	2.32	2.84	0.18	2.38–2.67	−0.27–0.18	0.21	0.677	**0.040**
		P	2.63	2.20	2.97	0.23	2.48–2.82				2.49	2.32	3.05	0.23	2.34–2.52				0.171
	AI	C	3.56	2.82	4.55	0.47	3.22–3.99	−0.58–0.05	0.72	**0.020**	3.49	2.93	3.81	0.27	3.37–3.70	−0.47–0.03	0.92	**0.040**	0.625
		P	3.25	2.45	4.17	0.39	2.94–3.39				3.27	3.03	3.69	0.19	3.20–3.35				0.807
	PMI	C	0.29	0.25	0.36	0.03	0.27–0.33	−0.02–0.01	0.10	0.718	0.26	0.22	0.33	0.03	0.23–0.29	−0.03–0.03	0.11	0.82	**0.039**
		P	0.29	0.22	0.37	0.03	0.27–0.31				0.26	0.24	0.32	0.02	0.24–0.28				**0.036**

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* effect size (rank-biserial correlation); *SI* symphysis index; *MI* mental index; *PI* posterior index; *AI* antegonial index; *PMI* panoramic mandibular index; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank test. $p < 0.05$ was considered statistically significant, and bold values indicate statistical significance

Table 6 Descriptive statistics and within-group comparisons of SI, MI, PI, AI, and PMI Values at baseline and 3-year follow-up in participants aged < 35 and ≥ 35 years

	< 35 years old										> 35 years old								
	Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
Initial	SI	C	2.30	1.67	3.23	0.47	1.93–2.58	−0.34–0.39	0.61	0.893	2.70	1.91	4.30	0.62	2.22–2.96	−0.43–0.27	0.15	0.645	0.082
		P	2.32	1.99	2.84	0.29	2.02–2.61				2.62	1.77	3.44	0.46	2.18–2.96				0.074
	MI	C	2.93	2.55	3.34	0.27	2.70–3.20	−0.14–0.34	0.38	0.402	3.27	2.09	5.01	0.61	2.85–3.58	−0.56–0.11	0.42	0.193	0.104
		P	3.03	2.67	3.46	0.25	2.81–3.23				3.05	2.31	3.74	0.43	2.67–3.55				0.895
	PI	C	2.48	2.12	2.77	0.21	2.31–2.70	−0.17–0.26	0.19	0.676	2.80	2.11	4.47	0.57	2.43–3.08	−0.33–0.25	0.08	0.780	0.091
		P	2.52	2.20	2.96	0.24	2.29–2.71				2.76	2.32	3.28	0.29	2.48–3.03				**0.034**
	AI	C	3.06	2.18	3.82	0.62	2.34–3.57	−0.46–0.45	0.01	0.982	3.90	2.64	5.84	0.74	3.49–4.31	−0.93–0.14	0.88	**0.080**	**0.040**
		P	3.05	2.55	3.53	0.29	2.88–3.30				3.36	2.54	4.27	0.44	3.05–3.75				**0.031**
	PMI	C	0.26	0.22	0.32	0.03	0.24–0.29	−0.02–0.03	0.22	0.617	0.31	0.20	0.45	0.06	0.26–0.36	−0.05–0.02	0.28	0.380	0.090
		P	0.27	0.22	0.31	0.03	0.25–0.29				0.29	0.23	0.36	0.04	0.26–0.33				**0.034**
Three years later	SI	C	2.39	2.06	2.91	0.29	2.10–2.67	−0.28–0.38	0.13	0.771	2.69	1.84	3.51	0.46	2.25–3.03	−0.42–0.06	0.47	0.147	0.074
		P	2.44	2.03	2.94	0.41	2.06–2.91				2.51	2.10	2.97	0.26	2.32–2.73				0.547
	MI	C	3.07	2.71	3.50	0.25	2.85–3.27	−0.52–0.05	1.17	**0.017**	3.05	2.35	3.78	0.41	2.71–3.35	−0.37–0.20	0.18	0.562	0.920
		P	2.78	2.55	3.19	0.24	2.57–3.04				2.97	2.09	3.71	0.49	2.62–3.34				0.263
	PI	C	2.52	2.20	2.96	0.25	2.29–2.71	−0.20–0.24	0.09	0.845	2.76	2.32	3.28	0.29	2.48–3.04	−0.31–0.02	0.53	0.101	**0.034**
		P	2.54	2.20	2.97	0.23	2.38–2.72				2.62	2.27	3.05	0.24	2.41–2.83				0.393
	AI	C	3.34	2.83	3.81	0.29	3.16–3.58	−0.57–0.16	0.52	**0.040**	3.64	2.82	4.55	0.45	3.34–4.03	−0.57–0.09	0.90	0.070	**0.034**
		P	3.14	2.45	4.17	0.47	2.90–3.29				3.31	2.82	3.73	0.26	3.17–3.48				0.195
	PMI	C	0.27	0.22	0.31	0.03	0.25–0.29	−0.03–0.01	0.39	0.394	0.29	0.23	0.36	0.04	0.26–0.34	−0.02–0.02	0.05	0.871	**0.027**
		P	0.26	0.22	0.29	0.03	0.24–0.28				0.29	0.24	0.37	0.03	0.27–0.31				**0.001**

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* effect size (rank-biserial correlation); *SI* symphysis index; *MI* mental index; *PI* posterior index; *AI* antegonial index; *PMI* panoramic mandibular index; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank test. $p < 0.05$ was considered statistically significant, and bold values indicate statistical significance

Table 7 Descriptive statistics and within-group comparisons of ROI1, ROI2, and ROI3 values at baseline and 3-year follow-up in female and male participants

	Female										Male								
	Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
Initial	ROI1	C	1.37	1.23	1.48	0.08	1.30–1.45	−0.14–0.01	0.69	**0.028**	1.33	0.99	1.47	0.15	1.30–1.44	−0.02–0.24	0.84	0.114	0.316
		P	1.29	0.90	1.52	0.14	1.23–1.40				1.44	1.23	1.55	0.10	1.37–1.50				**0.070**
	ROI2	C	1.44	1.34	1.55	0.06	1.40–1.48	−0.08–0.01	0.87	0.060	1.43	1.28	1.55	0.07	1.42–1.48	−0.09–0.03	0.45	0.388	0.817
		P	1.40	1.28	1.51	0.05	1.36–1.42				1.41	1.34	1.46	0.05	1.36–1.46				0.496
	ROI3	C	1.33	0.96	1.51	0.14	1.27–1.43	−0.11–0.07	0.17	0.578	1.35	1.12	1.47	0.12	1.29–1.44	−0.15–0.06	0.45	0.385	0.629
		P	1.30	1.02	1.52	0.15	1.19–1.45				1.31	1.18	1.41	0.08	1.23–1.40				0.870
Three years later	ROI1	C	1.34	1.03	1.51	0.12	1.25–1.43	−0.09–0.06	0.13	0.678	1.37	1.13	1.49	0.11	1.33–1.44	−0.08–0.14	0.33	0.527	0.479
		P	1.32	1.03	1.50	0.13	1.22–1.44				1.40	1.20	1.56	0.10	1.37–1.46				0.770
	ROI2	C	1.45	1.33	1.55	0.06	1.41–1.50	−0.09–0.02	1.17	**0.001**	1.43	1.32	1.59	0.08	1.40–1.46	−0.09–0.03	0.48	0.353	0.472
		P	1.39	1.32	1.48	0.04	1.36–1.42				1.40	1.37	1.47	0.03	1.38–1.43				0.453
	ROI3	C	1.31	0.72	1.53	0.17	1.23–1.42	−0.10–0.07	0.12	0.686	1.33	1.01	1.48	0.16	1.22–1.48	−0.16–0.13	0.09	0.866	0.749
		P	1.29	1.12	1.52	0.11	1.19–1.36				1.32	1.13	1.43	0.09	1.28–1.39				0.523

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* = effect size (rank-biserial correlation); *ROI1* condyle region; *ROI2* angular region; *ROI3* mental region; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank test. $p < 0.05$ was considered statistically significant, and bold values indicate statistical significance

Discussion

In this study, changes in mandibular bone structure over a 3-year follow-up were assessed in individuals diagnosed with PHPT and in age- and gender-matched healthy controls, using panoramic radiographs evaluated by radiomorphometric indices AI, MI, PMI, SI, PI, and FD analysis. Overall, the findings indicated no significant changes in radiomorphometric indices over 3 years; however, at baseline, the AI value was lower in the PHPT group than in controls, and this difference was statistically significant. In FD analysis, a significant/borderline difference at baseline was detected only for ROI2 (mandibular angle/angulus region); no significant differences were observed in the other ROIs or in changes over time. In subgroup analyses, the absence of differences between groups in age and gender distribution provided methodological strength for matching. Nevertheless, both radiomorphometric and FD data suggest that the effect of PHPT on mandibular cortical thinning and trabecular architectural changes may be limited but measurable.

In baseline panoramic radiographs, the AI value was 3.62 mm in the control group and 3.26 mm in the PHPT group, and the difference was statistically significant. At 3 years, AI values were 3.55 mm in the control group and 3.24 mm in the PHPT group; here, the difference was at the borderline level of statistical significance ($p \approx 0.050$) and should be interpreted cautiously in light of multiple comparisons. With respect to change over time, no marked decrease was observed in either group. The effect sizes (r) and confidence intervals reported in the tables suggest that these differences are small to moderate in magnitude.

In subgroup analyses, AI values in the PHPT group were lower than in controls in both female and male participants. By age group, the difference was more pronounced in participants younger than 35 years, whereas it did not reach statistical significance in those aged ≥ 35 years. It should be emphasized that these subgroup analyses are exploratory and that statistical power may be limited due to the small sample size.

In the literature, AI is defined as a parameter that provides information on bone mineral density by measuring the width of the mandibular cortex in the antegonial region, and many studies have reported that AI decreases in osteoporosis and metabolic bone diseases [31–34]. Aytekin et al. [19] reported significantly lower AI values in hyperthyroid patients than in controls (3.06 mm vs. 2.70 mm), and Miloğlu et al. [28] showed lower AI in PHPT patients than in healthy individuals (0.29 cm vs. 0.25 cm). Our findings are consistent with these data and support that AI may be a sensitive indicator reflecting cortical thinning in PHPT. Possible biological explanations include the acceleration of cortical resorption through increased osteoclastic activity driven by elevated PTH in PHPT, intense exposure of the antegonial region to masticatory forces, and age-related declines in cortical density.

At baseline, the MI value was 3.16 mm in the control group and 3.04 mm in the PHPT group; at 3 years, it measured 3.06 mm and 2.91 mm, respectively, with no statistically significant changes. By gender, no significant differences were detected in MI at baseline or at 3 years. In age subgroups, however, at the end of 3 years, the MI value in the PHPT group was significantly lower than in controls among participants younger than 35 years ($p = 0.017$), whereas no difference was observed in the ≥ 35 group. Considering studies reporting that MI is associated with low bone mineral density [35–37] and that MI decreases in osteoporosis [38, 39], it may be inferred that the effect of PHPT could emerge earlier and more distinctly in younger individuals; in contrast, physiological bone loss at older ages may mask the disease effect.

PMI was 0.29 in controls and 0.28 in PHPT at baseline; at 3 years, both groups remained at the same levels (control: 0.29; PHPT: 0.28), and the differences were not statistically significant. No significant differences were detected by gender or by age subgroup. While the literature reports that PMI can be used in osteoporosis screening [40–42], some studies have reported lower PMI in patient groups [19, 28, 31, 37, 43, 44]. In our sample, PMI appeared to have limited sensitivity for capturing PHPT-related cortical changes; potential reasons include small effect size, reduced sample size in subgroups, limited measurement precision for ROIs on panoramic radiographs, and a later manifestation of disease effects in PMI relative to other morphometric indices.

For SI and PI, baseline values were 2.52 mm and 2.69 mm in PHPT and 2.57 mm and 2.70 mm in controls; there was no significant change at 3 years, and between-group differences were not statistically significant. Similarly, no significant differences were observed in gender and age subgroups. Although these indices have been reported as auxiliary screening parameters in the literature [21, 45], the present study suggests that SI and PI may show limited sensitivity for detecting early PHPT-related changes in our sample.

FD analysis was performed in three regions: ROI1 (condyle), ROI2 (mandibular angle/angulus), and ROI3 (mental region). In overall analyses, baseline FD values were 1.47 in PHPT versus 1.48 in controls for ROI1; 1.50 in PHPT versus 1.53 in controls for ROI2; and 1.48 in PHPT versus 1.50 in controls for ROI3. At 3 years, no significant changes were observed in ROI1 and ROI3, whereas a slight decrease relative to baseline was noted in ROI2 in the PHPT group. The

Table 8 Descriptive statistics and within-group comparisons of ROI1, ROI2, and ROI3 values at baseline and 3-year follow-up in participants aged < 35 and ≥ 35 years

	< 35 years old										> 35 years old								
	Index	**Group**	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**Mean**	**Min**	**Max**	**SD**	**IQR**	**95% CI**	**r**	**P***	**P****
Initial	ROI1	C	1.38	1.29	1.45	0.06	1.34–1.45	−0.10–0.16	0.19	0.677	1.35	0.99	1.49	0.12	1.29–1.44	−0.12–0.01	0.51	0.112	0.462
		P	1.41	0.90	1.56	0.19	1.38–1.52				1.30	1.10	1.47	0.10	1.24–1.35				0.109
	ROI2	C	1.47	1.41	1.55	0.04	1.42–1.51	−0.10–0.02	1.44	0.050	1.43	1.29	1.56	0.06	1.38–1.47	−0.07–0.05	0.55	0.090	0.070
		P	1.41	1.35	1.49	0.04	1.38–1.43				1.39	1.29	1.51	0.06	1.35–1.42				0.533
	ROI3	C	1.37	1.05	1.51	0.13	1.28–1.47	−0.14–0.08	0.28	0.535	1.32	0.96	1.51	0.13	1.25–1.39	−0.12–0.06	0.20	0.530	0.308
		P	1.34	1.19	1.51	0.10	1.26–1.42				1.29	1.03	1.52	0.15	1.20–1.43				0.374
Three years later	ROI1	C	1.35	1.03	1.49	0.14	1.27–1.44	−0.07–0.16	0.34	0.456	1.34	1.13	1.52	0.11	1.26–1.45	−0.10–0.05	0.21	0.513	0.948
		P	1.39	1.18	1.57	0.11	1.32–1.46				1.32	1.03	1.46	0.13	1.21–1.46				0.147
	ROI2	C	1.47	1.40	1.59	0.06	1.42–1.51	−0.12–0.03	1.62	**0.002**	1.43	1.32	1.55	0.07	1.40–1.48	−0.07–0.02	0.71	**0.030**	0.213
		P	1.39	1.34	1.43	0.03	1.37–1.42				1.40	1.32	1.48	0.04	1.37–1.42				0.680
	ROI3	C	1.37	1.18	1.48	0.11	1.29–1.47	−0.19–0.05	0.99	**0.040**	1.29	0.72	1.54	0.18	1.23–1.41	−0.07–0.12	0.17	0.599	0.133
		P	1.27	1.13	1.44	0.09	1.19–1.34				1.32	1.13	1.52	0.12	1.22–1.40				0.319

C control; *P* patient; *Min* minimum; *Max* maximum; *SD* standard deviation; *IQR* interquartile range; *CI* confidence interval; *r* effect size (rank-biserial correlation); *ROI1* condyle region; *ROI2* angular region; *ROI3* mental region; * = Mann–Whitney *U* test; ** = Wilcoxon signed rank test. $p < 0.05$ was considered statistically significant, and bold values indicate statistical significance

group difference in ROI2 was borderline at baseline ($p = 0.050$) and significant at 3 years ($p < 0.01$); nevertheless, these findings should be interpreted with caution considering multiple comparisons. In gender subgroups, women showed higher baseline ROI2 FD in controls than in PHPT with a significant difference; in men, no ROI exhibited a significant difference. In age subgroups (< 35 and ≥ 35 years), ROI2 FD values tended to be lower in PHPT in both age groups, although statistical significance was not consistently reached. Cantürk et al. [27] reported a significant reduction in FD in the patient group at the mandibular angle region while observing no differences in other areas; our ROI2 finding is concordant in this respect. Dedeoğlu et al. [20] emphasized age-related declines in FD over five years; the younger mean age in our study and the 3-year follow-up may explain the absence of a pronounced global FD decrease. Biologically, the presence of both cortical and trabecular components in ROI2 and its intense exposure to masticatory forces may facilitate earlier detection of systemic metabolic effects in this region.

Methodologically, imaging parameters (device, exposure, positioning) can influence FD measurements; manual ROI selection also predisposes one to measurement variation. Although ROIs in this study were fixed in size and determined according to anatomical landmarks, small placement differences may still affect FD. Because the average of left and right values was used, potential lateralization effects could not be evaluated. Owing to the retrospective design, biochemical variables (serum Ca, PTH, phosphorus, ALP, etc.) could not be fully controlled, limiting direct linkage between radiographic findings and biological disease severity. Additionally, the lack of information on important confounding variables such as menopausal status, dietary factors, and socioeconomic background may have influenced mandibular bone characteristics and partially contributed to the observed variability. It should also be underscored that the subgroup analyses are exploratory and that statistical power may be limited due to sample size.

These findings suggest that panoramic radiographs in PHPT patients can serve not only dental treatment planning, but also as a supportive tool in screening systemic bone health. Clinically, combinations such as a low AI and decreased FD in ROI2 can be considered alerting signs that, in the presence of clinical suspicion, may prompt referral of patients for biochemical evaluation by relevant disciplines. Nonetheless, the integration of these parameters into routine screening and diagnostic algorithms will require larger, multicenter, prospective studies supported by biochemical data.

Conclusion

These findings indicate that, in this 3-year matched cohort, most radiomorphometric indices (SI, MI, PI, PMI) remained stable with no between-group differences, while AI was consistently lower in PHPT and fractal differences were confined to the mandibular angle (ROI2). Clinically, the combination of a low AI and reduced ROI2 FD on routine panoramic radiographs may serve as an adjunctive red flag—when clinical suspicion exists—prompting referral for biochemical evaluation, rather than functioning as standalone diagnostic criteria. Given the small effect sizes, the borderline significance at baseline, and multiple comparisons, integration of these metrics into screening or diagnostic algorithms should await confirmation in larger, multicenter, prospective studies that include contemporaneous biochemical profiling and reproducibility assessments.

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Author contributions
FO and DC data collection and management, data analysis, manuscript writing and played a role in design of the study. YY and AP data analysis, manuscript writing and played a role in design of the study.

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Data availability
The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was conducted in accordance with the principles of the Declaration of Helsinki. Permission for the study was obtained from the Ordu University Clinical Research Ethics Committee with the decision number 2024/178.

Informed consent
Informed consent was obtained from all participants included in the study.

Competing interests
The authors declare no competing interests.

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References

1. Fraser WD. Hyperparathyroidism. Lancet. 2009;374(9684):145–58.
2. Tassone F, Gianotti L, Baffoni C, Visconti G, Pellegrino M, Cassibba S, et al. Vitamin D status in primary hyperparathyroidism: a Southern European perspective. Clin Endocrinol (Oxf). 2013;79(6):784–90.

3. Palermo A, Tabacco G, Makras P, Zavatta G, Trimboli P, Castellano E, et al. Primary hyperparathyroidism: from guidelines to outpatient clinic. Rev Endocr Metab Disord. 2024;25(5):875–96.
4. Rolighed L, Rejnmark L, Christiansen P. Bone involvement in primary hyperparathyroidism and changes after parathyroidectomy. Eur Endocrinol. 2014;10(1):84–7.
5. Silva BC, Bilezikian JP. Parathyroid hormone: anabolic and catabolic actions on the skeleton. Curr Opin Pharmacol. 2015;22:41–50.
6. Guimarães LM, Valeriano AT, Rebelo Pontes HA, Gomez RS, Gomes CC. Manifestations of hyperparathyroidism in the jaws: concepts, mechanisms, and clinical aspects. Oral Surg Oral Med Oral Pathol Oral Radiol. 2022;133(5):547–55.
7. Mittal S, Gupta D, Sekhri S, Goyal S. Oral manifestations of parathyroid disorders and its dental management. J Dent Allied Sci. 2014;3(1):34–8.
8. Ibrahem HM. Gnathic bones and hyperparathyroidism: a review on the metabolic bony changes affecting the mandible and maxilla in case of hyperparathyroidism. Adv Med. 2020;2020(1):6836123.
9. Lajolo C, Patini R, Limongelli L, Favia G, Tempesta A, Contaldo M, et al. Brown tumors of the oral cavity: presentation of 4 new cases and a systematic literature review. Oral Surg Oral Med Oral Pathol Oral Radiol. 2020;129(6):575-84.e4.
10. Olsen J, Sealey C. Brown tumour of the mandible in primary hyperparathyroidism; a case report. N Z Dent J. 2015;111(3):1.
11. Brabyn P, Capote A, Belloti M, Zylberberg I. Hyperparathyroidism diagnosed due to brown tumors of the jaw: a case report and literature review. J Oral Maxillofac Surg. 2017;75(10):2162–9.
12. Manjunatha BS, Purohit S, Harsh A, Vangala N. A complex case of brown tumors as initial manifestation of primary hyperparathyroidism in a young female. J Oral Maxillofac Pathol. 2019;23(3):477.
13. Raubenheimer EJ, Noffke CE, Mohamed A. Expansive jaw lesions in chronic kidney disease: review of the literature and a report of two cases. Oral Surg Oral Med Oral Pathol Oral Radiol. 2015;119(3):340–5.
14. Chang JI, Som PM, Lawson W. Unique imaging findings in the facial bones of renal osteodystrophy. AJNR Am J Neuroradiol. 2007;28(4):608–9.
15. Palla B, Burian E, Fliefel R, Otto S. Systematic review of oral manifestations related to hyperparathyroidism. Clin Oral Investig. 2018;22(1):1–27.
16. Shah N, Bansal N, Logani A. Recent advances in imaging technologies in dentistry. World J Radiol. 2014;6(10):794–807.
17. Izzetti R, Nisi M, Aringhieri G, Crocetti L, Graziani F, Nardi C. Basic knowledge and new advances in panoramic radiography imaging techniques: a narrative review on what dentists and radiologists should know. Appl Sci. 2021;11(17):7858.
18. Vlasiadis KZ, Damilakis J, Velegrakis GA, Skouteris CA, Fragouli I, Goumenou A, et al. Relationship between BMD, dental panoramic radiographic findings and biochemical markers of bone turnover in diagnosis of osteoporosis. Maturitas. 2008;59(3):226–33.
19. Aytekin Z, Yilmaz SG. Evaluation of osseous changes in dental panoramic radiography using radiomorphometric indices in patients with hyperthyroidism. Oral Surg Oral Med Oral Pathol Oral Radiol. 2022;133(4):492–9.
20. Dedeoğlu N, Eşer G, Çelik Özen D, Altun O. Five-year change of panoramic radiomorphometric indices and fractal dimension values in type 2 diabetes patients. Oral Radiol. 2024;40(1):49–57.
21. Gunacar DN, Goller Bulut D, Ustaoglu G, Yildirim EA. In vivo evaluation of mandibular trabecular and cortical bone quality in psoriasis vulgaris. Oral Surg Oral Med Oral Pathol Oral Radiol. 2022;133(5):604–11.
22. Kavitha MS, Park SY, Heo MS, Chien SI. Distributional variations in the quantitative cortical and trabecular bone radiographic measurements of mandible, between male and female populations of Korea, and its utilization. PLoS ONE. 2016;11(12):e0167992.
23. Mandelbrot BB. The fractal geometry of nature. New York: Macmillan, W. H Freeman and Co; 1982. p. 170.
24. Updike SX, Nowzari H. Fractal analysis of dental radiographs to detect periodontitis-induced trabecular changes. J Periodontal Res. 2008;43(6):658–64.
25. Gumussoy I, Miloglu O, Cankaya E, Bayrakdar IS. Fractal properties of the trabecular pattern of the mandible in chronic renal failure. Dentomaxillofac Radiol. 2016;45(5):20150389.
26. Mohammadi M, Alinejad S, Abesi F. Jaw bone changes in panoramic radiography in patients with hyperparathyroidism. J Shahrekord Univ Med Sci. 2024;26(3):101–7.
27. Canturk F, Miloglu O, Gumussoy I, Dayanan R. The investigation of radiographic findings of mandibles on panoramic radiographs of patients with primary hyperparathyroidism using fractal analysis. Eur J Med. 2021;53(3):185.
28. Miloğlu Ö, Cantürk F, Akkaya G, Bilen A. The investigation of osteoporotic changes in patients with primary hyperparathyroidism with radiomorphometric indices determined on panoramic radiography. Turk Klin J Dent Sci. 2021;27(4):1.
29. Ergun S, Saraçoglu A, Guneri P, Ozpınar B. Application of fractal analysis in hyperparathyroidism. Dentomaxillofac Radiol. 2009;38(5):281–9.
30. White SC, Rudolph DJ. Alterations of the trabecular pattern of the jaws in patients with osteoporosis. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 1999;88(5):628–35.
31. Karga H, Papapetrou PD, Korakovouni A, Papandroulaki F, Polymeris A, Pampouras G. Bone mineral density in hyperthyroidism. Clin Endocrinol. 2004;61(4):466–72.
32. Dagistan S, Bilge O. Comparison of antegonial index, mental index, panoramic mandibular index and mandibular cortical index values in the panoramic radiographs of normal males and male patients with osteoporosis. Dentomaxillofac Radiol. 2010;39(5):290–4.
33. Abdinian M, Salehi MM, Mortazavi M, Salehi H, Kazemi Naeini M. Comparison of dental and skeletal indices between patients under haemodialysis and peritoneal dialysis with healthy individuals in digital panoramic radiography. Dentomaxillofac Radiol. 2021;50(1):20200108.
34. Mahl CRW, Licks R, Fontanella VRC. Comparison of morphometric indices obtained from dental panoramic radiography for identifying individuals with osteoporosis/osteopenia. Radiol Bras. 2008;41:183–7.
35. Dutra V, Yang J, Devlin H, Susin C. Radiomorphometric indices and their relation to gender, age, and dental status. Oral Surg Oral Med Oral Pathol Oral Radiol Endod. 2005;99(4):479–84.
36. Muramatsu C, Horiba K, Hayashi T, Fukui T, Hara T, Katsumata A, et al. Quantitative assessment of mandibular cortical erosion on dental panoramic radiographs for screening osteoporosis. Int J Comput Assist Radiol Surg. 2016;11:2021–32.
37. Kwon A-Y, Huh K-H, Yi W-J, Lee S-S, Choi S-C, Heo M-S. Is the panoramic mandibular index useful for bone quality evaluation? Imaging Sci Dent. 2017;47(2):87.
38. Yalcin ED, Avcu N, Uysal S, Arslan U. Evaluation of radiomorphometric indices and bone findings on panoramic images in patients with scleroderma. Oral Surg Oral Med Oral Pathol Oral Radiol. 2019;127(1):e23–30.
39. Waghray S, Velpula N, Dara BGB, Duddu MK, Narayen V, Allam NSJ. Mandibulo-osseous predictors of osteoporosis: a double-blind study on the correlation and comparison of mental index with bone mineral density in post-menopausal women. J Indian Acad Oral Med Radiol. 2015;27(2):183–8.
40. Limeira FIR, Rebouças PRM, Diniz DN, Melo DPD, Bento PM. Decrease in mandibular cortical in patients with type 1 diabetes mellitus combined with poor glycemic control. Braz Dental J. 2017;28(5):552–8.
41. Benson BW, Prihoda TJ, Glass BJ. Variations in adult cortical bone mass as measured by a panoramic mandibular index. Oral Surg Oral Med Oral Pathol. 1991;71(3):349–56.
42. Klemetti E, Kolmakov S, Kröger H. Pantomography in assessment of the osteoporosis risk group. Eur J Oral Sci. 1994;102(1):68–72.
43. Devlin H, Horner K. Mandibular radiomorphometric indices in the diagnosis of reduced skeletal bone mineral density. Osteoporos Int. 2002;13:373–8.
44. Ledgerton D, Horner K, Devlin H, Worthington H. Radiomorphometric indices of the mandible in a British female population. Dentomaxillofac Radiol. 1999;28(3):173–81.
45. Ustaoğlu G, Göller Bulut D, Bayrak S. Applicability of dental panoramic radiomorphometric and fractal dimension analysis in the evaluation of antiepileptic drug-induced osteoporosis. Int J Med Dent. 2019;23(2):1.

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